SHEETLINK: a SAS/R helper library for driving Smartsheet (no AI)

What it is. A small, deterministic library that lets a biostatistician keep a Smartsheet program tracker current **automatically from scheduled SAS or R**, no copy-paste, no manual status updates, no LLM. SAS/R writes the *truthful operational status* to Smartsheet by a stable key; **Smartsheet's own built-in workflows send the notifications.** That split is the whole design: *code owns the data*, *Smartsheet owns the alerts*.

```
validated nightly extract —▶ SAS/R (this library) —▶ Smartsheet cells (by key)
                                  |
                                  Smartsheet Automation (configured once)
                                  ▼
                                  email / mobile push / Slack / Teams
```

Files

File	Role
`ss_macros.sas`	The SAS library. Every macro is `%ss_*`. PROC HTTP + JSON LIBNAME engine.
`ss_companion.R`	The R equivalent (httr2 + jsonlite). Same governance, same idempotency.
`ss_config.sas`	Per-sheet constants: sheet id, key column, recipients, **the ops-only allowlist**.
`tracker_update.sas`	The nightly driver you schedule. Build data → `%ss_upsert` → (attach) → heartbeat.

The five things this library guarantees

1. **Idempotent, never duplicates.** `%ss_upsert` / `ss_upsert()` match on a **stable external key you own** (a Task/Deliverable/Milestone code). If the key is on the sheet it **updates that row in place** (PUT); if not, it **appends** (POST). Run it ten times a night: the sheet converges to the same state. *Never blind-append.*
2. **Ops-only, no PHI ever leaves the tenant.** `%ss_guard` / `ss_guard()` check **every column you are about to write against an allowlist** (`CFG_ALLOW`). Try to push a non-allowlisted column and the job **fails loud and stops**: a coded boundary, not a habit. Allowed: milestone dates, deliverable/QC status, % complete, aggregate counts, ownership. **Never:** participant-level, unblinded, PHI, or reported clinical numbers.
3. **Token is never hard-coded or logged.** Read at runtime from `SMARTSHEET_TOKEN` (or a

perms-600 file), inside `OPTIONS NOSOURCE` so it never reaches the log. Rotate it on the Smartsheet side; the code never changes.

4. **Rate-limit aware.** Smartsheet allows ~300 requests/min/token. `%ss_http` / `req_retry()` retry **429 and 5xx with exponential backoff** and otherwise **fail loud**.
5. **Fails loud, never silent.** A stale tracker that *looks* fresh is the dangerous failure. `%ss_assert` emails `CFG_BACKUP` and aborts on any non-2xx, and the driver ends with a heartbeat so silence is unambiguously a failure.

Newer helpers (SAS · R · PowerShell)

- **Dry run: validate before you schedule.** `ss_upsert(..., dryrun = TRUE)` (R), `%ss_upsert(..., dryrun=Y)` (SAS), `-DryRun` (PowerShell) compute exactly what *would* be updated and added and print the counts, **without sending anything**. Run it once by hand before you put the job on a scheduler.
- **Read-back / verification.** `ss_get(sheet, key, keycol, col)` (R) and `%ss_get(sheet=, key=, keycolname=, col=, mvar=)` (SAS) return one cell's current value by its key: handy for a read-after-write check or a guard condition.
- **Empty means "leave it."** A missing/NA/blank value for a column is now **omitted** from the write in all three (it was already so in SAS): the cell is left untouched rather than overwritten with a blank or the literal text `NA`.

See it / run it, no Smartsheet account needed

`../_demo/` runs this **exact** library end-to-end against a local stand-in of the Smartsheet REST API, so you can watch the real HTTP calls, the idempotent upsert, the attach, and the allowlist guard actually happen:

```
node ../_demo/ss_mock_api.js &                    # representative Smartsheet API (localhost)
SMARTSHEET_TOKEN=demo CP101_TRACKER_ID=5500 Rscript ../_demo/ss_run_demo.R
```

Or, with no terminal at all, open **`../CP101_Tracker_Demo.html`** (the interactive tracker). The captured output of the run above is in `../_demo/capture/`. The SAS macros have not been executed (no SAS license).

Quick start (SAS)

```sas
/ once: export your token in the scheduler's environment, NOT in code /
/   export SMARTSHEET_TOKEN=xxxxxxxx   (or store in a perms-600 file)  /
%include "ss_config.sas";   / set CFG_SHEET, CFG_KEYCOLNAME, CFG_ALLOW ... /
```

```
%include "ss_macros.sas";
%ss_init(tokenenv=&CFG_TOKEN_ENV, backup=&CFG_BACKUP, allow=&CFG_ALLOW);
data tracker;
length TaskID $40 Status $20;
TaskID="CP101-DBL"; Status="At Risk"; PctDone=0.60; output;
label Status="Status" PctDone="% Done";   / each label = the Smartsheet column TITLE /
run;
%ss_upsert(sheet=&CFG_SHEET, keycol=TaskID, keycolname=&CFG_KEYCOLNAME, data=tracker);
```

The **contract for `%ss_upsert`**: your dataset has the key variable plus one variable per column to write, and **each value variable is labeled with its exact Smartsheet column title** (`label Status="Status" PctDone="% Done";`). The macro reads those labels, resolves each to a `columnId` from a freshly-fetched column map, JSON-escapes values, and splits the work into a PUT (existing keys) and a POST (new keys).

Quick start (R)

```
source("ss_companion.R")
allow <- c("Task ID","Status","% Done","Owner","Due")
df <- tibble::tibble(`Task ID` = c("CP101-DBL","CP101-ADaM"),
Status    = c("At Risk","On Track"),
`% Done` = c(0.60, 0.85))
ss_upsert(sheet = Sys.getenv("CP101_TRACKER_ID"), key = "Task ID", df = df, allow = allow)
```

In R the **data-frame column names are the Smartsheet titles** (back-tick names with spaces).

Macro / function reference

SAS	R	Does
`%ss_init`	`ss_req` (implicit)	Load token (hidden), base URL, allowlist, fail-loud target.
`%ss_http`	`ss_req`	One authenticated call; 429/5xx retry + backoff; fail loud on non-2xx.
`%ss_colmap`	`ss_colmap`	GET sheet → fresh **column title → columnId** map (rebuilt every run; ids are the only stable address).
`%ss_get_rows`	`ss_get_rows`	GET sheet rows → long (rowId, columnId, value); the key→rowId map.

SAS	R	Does
`%ss_guard`	`ss_guard`	Enforce the **ops-only allowlist**; fail loud on any non-allowlisted column.
`%ss_upsert`	`ss_upsert`	**Idempotent** update-by-key / add-if-new. The flagship.
`%ss_attach`	`ss_attach`	Attach an ops-only status PDF to a row (rate-intensive; once per run).
`%ss_update_request`	`ss_update_request`	Supported human-in-the-loop ask (emails a link to update rows).
`%ss_assert`	`ss_assert`	Fail loud + notify backup + abort.

Notifications: configure once in Smartsheet, not in code (Pattern A)

The robust pattern is **not** to send emails from SAS. It is:

1. SAS/R writes the truthful cell (e.g. `Status = "At Risk"`).
2. A Smartsheet **Automation** you set up once, *When `Status` changes to `At Risk` → Alert the PM / send a mobile push / post to a Teams channel*: fires.

This keeps recipients, escalation, quiet-hours, and channels in Smartsheet (where a PM can edit them) and keeps your code about *data*. Use `%ss_update_request` only for a targeted "please confirm X" ask that needs a specific person to act.

Governance (the boundary that lets this be automated at all)

- **Smartsheet is a tracker, not a system of record.** The validated SAS/R outputs remain the source of truth; the sheet is a convenience mirror of *operational* status.
- **Allowlist is code, not a guideline.** Widening it is a reviewed change to `CFG_ALLOW`.
- **Aggregate only.** Counts and statuses, never a value attributable to one participant.
- Token in env/secret store, rotated, never logged. Least-privilege service account.

A note on robustness

The JSON LIBNAME engine names its child tables from the response structure (`COLUMNS`, `ROWS`, `ROWS_CELLS`). If your SAS version names them differently, run a GET once and `PROC DATASETS lib=_ssj;` to confirm, then adjust `%ss_colmap`/`%ss_get_rows`.

Make the key a TEXT column. The idempotent match is on the key value as text on both sides (`cats()`), so the Smartsheet key column should be a **Text/Number column holding text codes** (`CP101-DBL`), not an auto-number or a numeric column, otherwise a zero-padded code (`007`) can fail to match its own prior row and a duplicate is appended. Use a stable code you own as the key.

Validate the library against a **non-production sandbox sheet** before scheduling it against a

live program tracker: the same way you'd validate any macro that writes to a shared artifact.